

# Shuo Liu

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## Education

### University of Washington

Sep. 2024 – Mar. 2026

*M.S. in Electrical and Computer Engineering*

- GPA: 3.99/4.00
- Research Interest: Robot Learning
- Advisor: Prof. Ranjay Krishna, Dr. Jiafei Duan

### Southeast University

Sep. 2020 – Jun. 2024

*B.E. in Robot Engineering*

- GPA: 87.5/100
- Research Interest: 3D reconstruction and SLAM
- Advisor: Prof. Chenxing Wang

## Publications

\* equal contribution    ‡ equal advising    § core contributors

### [1] MolmoAct2: Action Reasoning Models for Real-World Deployment [\[Arxiv\]](#) [\[Website\]](#)

Haoquan Fang<sup>\*§</sup>, Jiafei Duan<sup>\*§</sup>, Donovan Clay<sup>§</sup>, Sam Wang<sup>§</sup>, **Shuo Liu<sup>§</sup>**, Weikai Huang<sup>§</sup>, Xiang Fan<sup>§</sup>, Wei-Chuan Tsai<sup>§</sup>, Shirui Chen<sup>§</sup>, Yi Ru Wang<sup>§</sup>, Shanli Xing<sup>§</sup>, Jaemin Cho, Jae Sung Park, Ainaz Eftekhari, Peter Sushko, Karen Farley, Angad Wadhwa, Cole Harrison, Winson Han, Ying-Chun Lee, Eli Vanderbilt, Rose Hendrix, Suveen Ellawela, Lucas Ngoo, Joyce Chai, Zhongzheng Ren<sup>§</sup>, Ali Farhadi<sup>§</sup>, Dieter Fox<sup>§</sup>, Ranjay Krishna<sup>§</sup>.

### [2] WildDet3D: Scaling Promptable 3D Detection in the Wild [\[Arxiv\]](#) [\[Website\]](#)

Weikai Huang<sup>§</sup>, Jieyu Zhang<sup>§</sup>, Sijun Li, Taoyang Jia, Jiafei Duan, Yunqian Cheng, Jaemin Cho, Matthew Wallingford, Rustin Soraki, Chris Dongjoo Kim, **Shuo Liu**, Donovan Clay, Taira Anderson, Winson Han, Ali Farhadi, Bharath Hariharan, Zhongzheng Ren<sup>§</sup>, Ranjay Krishna<sup>§</sup>.

*Under review*

### [3] VLS: Steering Pretrained Robot Policies via Vision–Language Models [\[Arxiv\]](#) [\[Website\]](#)

**Shuo Liu**, Ishneet Sukhvinder Singh, Yiqing Xu, Jiafei Duan<sup>‡</sup>, Ranjay Krishna<sup>‡</sup>

*Under review*

### [4] MolmoSpaces: Large-Scale Open Ecosystem for Robot Manipulation and Navigation [\[Arxiv\]](#) [\[Website\]](#)

Yejin Kim<sup>\*§</sup>, Wilbert Pumacay<sup>\*§</sup>, Omar Rayyan<sup>\*§</sup>, Max Argus<sup>\*§</sup>, Winson Han<sup>§</sup>, Eli Vanderbilt<sup>§</sup>, Jordi Salvador<sup>§</sup>, Abhay Deshpande<sup>§</sup>, Rose Hendrix<sup>§</sup>, Snehal Jauhri<sup>§</sup>, **Shuo Liu<sup>§</sup>**, Nur Muhammad Mahi Shafiullah, Maya Guru, Ainaz Eftekhari, Karen Farley, Donovan Clay, Jiafei Duan, Arjun Guru, Piper Wolters, Alvaro Herrasti, Ying-Chun Lee, Georgia Chalvatzaki, Yuchen Cui, Ali Farhadi, Dieter Fox, Ranjay Krishna<sup>§</sup>.

*Robotics: Science and Systems (RSS), 2026*

### [5] MolmoAct: Action Reasoning Models that can Reason in Space [\[Arxiv\]](#) [\[Website\]](#)

Jason Lee<sup>\*§</sup>, Jiafei Duan<sup>\*§</sup>, Haoquan Fang<sup>\*§</sup>, Yuquan Deng<sup>§</sup>, **Shuo Liu<sup>§</sup>**, Boyang Li<sup>§</sup>, Bohan Fang<sup>§</sup>, Jieyu Zhang<sup>§</sup>, Yi Ru Wang<sup>§</sup>, Sangho Lee, Winson Han, Wilbert Pumacay, Angelica Wu, Rose Hendrix, Karen Farley, Eli Vanderbilt, Ali Farhadi, Dieter Fox<sup>§</sup>, Ranjay Krishna<sup>§</sup>.

*IEEE International Conference on Robotics and Automation (ICRA), 2026*

*Conference on Robot Learning (CoRL) 2025 workshop on Making Sense of Data in Robotics*

*Conference on Robot Learning (CoRL) 2025 workshop on Resource-Rational Robot Learning*

### [6] FORGE-Tree: Diffusion-Forcing Tree Search for Long-Horizon Robot Manipulation [\[Arxiv\]](#) [\[Website\]](#)

Yanjia Huang<sup>\*</sup>, **Shuo Liu<sup>\*</sup>**, Sheng Liu, Qingxiao Xu, Mingyang Wu, Xiangbo Gao, Zhengzhong Tu.

*Under review*

## Research Experience

**Student Researcher**, Allen Institute for Artificial Intelligence

Nov. 2025 – Mar. 2026

**Student Collaborator**, Allen Institute for Artificial Intelligence

Apr. 2025 – Oct. 2025

*Perceptual Reasoning and Interaction Research (PRIOR)*

*Advised by Prof. Ranjay Krishna*

- Led the VLS project, developing the complete system pipeline and performing evaluations.

- Contributed to the MolmoSpaces navigation benchmark, including task design and experimental evaluation.
- Built standardized data collection and evaluation pipelines for MolmoAct experiments on the RBY1 robot.

**Research Collaborator**, Texas A&M University

Nov. 2024 – Mar. 2025

*Trustworthy, Autonomous, Human-Centered, and Embodied Intelligence Group*

*Advised by Prof. Zhengzhong Tu*

- Contributed to FORGE-Tree project for method design and paper writing.

**Research Engineer Intern**, Hangzhou Qiancen Intelligent Technology Co., Ltd.

Mar. 2024 – Aug. 2024

- 3D reconstruction of autonomous driving scenes based on 3D Gaussian Splatting.

## **Skills**

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**Programming:** C, C++, Python, MATLAB

**Tools:** PyTorch, Lerobot, Isaac Sim/Lab, ManiSkill, MuJoCo, PyBullet, ROS, RLBench

**Robots:** Franka Emika Panda/Research 3, Rainbow RBY-1, Ufactory X-arm7, Ninova Gen 3